

Pillar 5: Capstone Project

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Post Graduate Diploma in Artificial Intelligence and Machine Learning

PHASE 07 | FetalGuard AI (CTG — Fetal Risk Classifier) — Deployment App

A Streamlit clinical decision-support tool for the cost-sensitive CTG abnormality classifier built across Phases 1–5.

Important Notes

- Use files in folder '02 src' for demonstration

Setup.

```
pip install -r requirements.txt
```

1. Build the deployment artifact (run once, or after retraining)

This loads `ctg_preprocessed.pkl` (produced by the Phase 02 notebook), trains the Phase 03 best model (XGBoost + `scale_pos_weight`), computes SHAP feature importance, and saves everything the app needs into `deployment_artifact.pkl`.

```
python build_artifact.py
```

Requires `ctg_preprocessed.pkl` to be present in this directory (copy it over from the Phase 02 notebook's working directory).

2. Run the app

```
streamlit run app.py
```

What the app does

- **Inputs:** sliders for the top-5 SHAP-ranked features (`ASTV` , `AC` , `Variance` , `ALTV` , `Width`) on this dataset — recomputed automatically if the model changes).

- **Logic:** builds a full 22-feature vector (other features default to training-set medians), scales it with the *saved* Phase 02 scaler (never refit), predicts probability, and applies the configurable threshold.
- **Output:** Normal / High Risk label, a confidence gauge, and a natural-language SHAP-based explanation of the top contributing factors for *that specific prediction* (not just global importance).
- **Sensitivity Slider:** lets the user move the decision threshold away from the cost-optimal default (0.10) to see how the same patient's classification would change under a more or less conservative policy.

Files

File	Purpose
<code>app.py</code>	The Streamlit application
<code>build_artifact.py</code>	Builds <code>deployment_artifact.pkl</code> from Phase 02 output
<code>genai_tools.py</code>	Generative AI interface to improve usability, interpretation, and reporting
<code>deployment_artifact.pkl</code>	Model + scaler + feature stats + SHAP metadata (generated)
<code>requirements.txt</code>	Python dependencies

Important caveat

This UI exposes only 5 of the model's 22 input features for usability. The remaining 17 are held at training-set medians. This means predictions from this simplified form will **not exactly match** what the full model would predict given a complete real CTG trace — it's a demonstration/triage tool, not a replacement for the full clinical pipeline. This tradeoff is disclosed in-app via an info banner.